

Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor

Professional Project Report	Data Analytics / Supply Chain Analysis
Prepared by	Nikhil Chikte
Project Status	Analysis, dashboard and cloud deployment completed
Dashboard	Interactive Streamlit application
Repository	nassau-candy-shipping-analysis

Final portfolio edition updated after evaluator feedback. The dashboard now includes dynamic KPI context, date/region/shipping-mode filters, reset controls, monthly trends, regional rankings, route contribution percentages, and improved visual storytelling.

1. Executive Summary

This project evaluates factory-to-customer shipping activity for Nassau Candy Distributor using order, shipment, customer, product, sales, profitability, and geographic data. The final dataset contains 10,194 shipment records. The analysis combines data validation, feature engineering, KPI measurement, regional and shipping-mode comparison, route ranking, monthly trend analysis, and an interactive Streamlit dashboard.

Pacific is the largest shipment region and Standard Class is the dominant shipping mode. The Chocolate to California route is the highest-volume route. A major limitation remains the unusually high calculated shipping lead time, which should be treated as a data-quality signal until the source dates are validated.

2. Project Objectives

- Analyze factory-to-customer shipment activity and shipping lead times.
- Compare regional performance using shipment volume, sales, units, lead time, share, and rank.
- Evaluate Standard Class, Second Class, First Class, and Same Day shipping modes.
- Identify and rank high-volume factory-to-customer routes.
- Analyze monthly sales and shipment trends.
- Build an interactive dashboard with filters, reset controls, dynamic KPIs, charts, tables, and business insights.
- Document data-quality limitations and avoid unsupported operational conclusions.

3. Dataset & Analytical Fields

The dashboard covers records from approximately January 2024 through December 2025. Key fields include Order ID, Order Date, Ship Date, Ship Mode, Customer ID, City, State/Province, Region, Product ID, Product Name, Sales, Units, Gross Profit, Cost, Factory Region, Factory State, Shipping Lead Time, and Factory-to-Customer Route.

Category	Representative fields
Order / Shipment	Order ID, Order Date, Ship Date, Ship Mode
Customer / Geography	Customer ID, City, State/Province, Region
Product / Financial	Product ID, Product Name, Sales, Units, Gross Profit, Cost
Route Features	Factory Region, Factory State, Shipping Lead Time, Factory-to-Customer Route

4. Methodology

4.1 Data Cleaning & Validation

Dataset structure, data types, missing values, duplicate records, date fields, and lead-time validity were reviewed before analysis.

4.2 Feature Engineering

Shipping Lead Time = Ship Date - Order Date. Factory and customer geographic attributes were combined to support route-level analysis.

4.3 Aggregation & Benchmarking

Records were aggregated by region, shipping mode, month, and factory-to-customer route. Rankings and percentage contributions were added to improve comparative interpretation.

4.4 Interactive Dashboard

The final Streamlit application supports region, shipping-mode, and date-range filtering. KPIs, tables, charts, route rankings, and insights update dynamically. A Reset Filters control and filtered-record count improve usability.

5. Key Performance Indicators

KPI	Overall result
Total Shipments	10,194
Total Sales	\$141,783.63
Total Profit	\$93,442.80
Total Units	38,654
Average Calculated Lead Time	1,320.84 days

Interpretation: The average lead-time value is exceptionally high for normal operational shipping. It is retained for analytical comparison but should not be treated as a verified service-level benchmark.

6. Regional Performance & Ranking

Rank	Region	Shipments	Shipment Share	Sales
1	Pacific	3,253	31.91%	\$46,301.53
2	Atlantic	2,986	29.29%	\$41,197.24
3	Interior	2,335	22.91%	\$32,037.60
4	Gulf	1,620	15.89%	\$22,247.26

Pacific ranks first with 3,253 shipments (31.91% of all records). Atlantic ranks second. Together, Pacific and Atlantic represent about 61.2% of shipment activity, making them important areas for monitoring.

7. Shipping Mode Performance

Shipping Mode	Shipments	Share
Standard Class	6,120	60.04%
Second Class	1,979	19.41%
First Class	1,548	15.19%
Same Day	547	5.37%

Standard Class is the dominant shipping mode, representing approximately 60% of all shipment records. The dashboard highlights the leading mode while allowing comparisons to change dynamically with filters.

8. Factory-to-Customer Route Performance

Rank	Route	Shipments	Share
1	Chocolate -> California	1,948	19.1%
2	Chocolate -> New York	1,081	10.6%
3	Chocolate -> Texas	957	9.4%
4	Chocolate -> Pennsylvania	558	5.5%
5	Chocolate -> Washington	490	4.8%

Chocolate -> California is the highest-volume route with 1,948 shipments, approximately 19.1% of the dataset. The final dashboard presents a Top 10 route chart with shipment counts and contribution percentages.

9. Monthly Trend Analysis

The final dashboard adds monthly Sales and Shipment trends. These views make it easier to identify changes over time and visually highlight peak periods. The trend charts use a consistent dark visual theme with blue series and orange peak markers.

10. Dashboard Improvements After Evaluation

- Added KPI comparisons and contextual percentage information.
- Added filtered-record count and percentage.
- Added a Reset Filters control.
- Added monthly sales and shipment trend charts.
- Added regional shipment-share percentages and ranking.
- Improved filter consistency across KPIs, tables, charts, routes, and insights.
- Added route contribution percentages.
- Improved chart styling, hierarchy, highlighting, and visual storytelling.

These changes directly address the evaluator's usability and storytelling feedback and make the dashboard more suitable for portfolio presentation.

11. Key Business Insights

- Pacific is the highest-volume region with 3,253 shipments and approximately 31.9% of total shipment activity.
- Atlantic ranks second with 2,986 shipments.
- Pacific generates approximately \$46,301.53 in sales, the highest among the four regions.
- Standard Class is the most frequently used shipping mode with 6,120 shipments.
- Same Day has the lowest shipment volume with 547 shipments.
- Chocolate -> California is the highest-volume factory-to-customer route with 1,948 shipments.
- Monthly trend views support identification of changing and peak shipment/sales periods.
- Regional rankings and contribution percentages make geographic comparisons easier to interpret.

12. Data Quality & Limitations

Calculated shipping lead times are unusually high. The overall average is approximately 1,320.84 days, and observed minimum values are around 904 days. These values are inconsistent with normal operational shipping timelines.

Possible explanations include historical or simulated records, transformed source data, mismatched date fields, or other source-data quality issues. Order Date and Ship Date should therefore be validated against original business records before operational decisions are made.

13. Business Recommendations

- Validate Order Date and Ship Date before using lead-time metrics for service-level decisions.
- Prioritize high-volume routes for deeper operational investigation.
- Monitor Pacific and Atlantic closely because they represent the majority of shipment activity.
- Analyze premium modes such as Same Day and First Class separately when service-level data is validated.
- Add geographic distance and shipping-cost data in future versions for true route-efficiency analysis.
- Use the current dashboard primarily as an analytical monitoring and comparison framework.

14. Business Value

The project provides a reusable framework for logistics monitoring. It helps identify high-volume routes, compare regions, understand shipping-mode utilization, track trends, rank operational activity, and prioritize areas for further investigation.

15. Tools & Technologies

Tool	Purpose
Python	Core analysis and application development
Pandas	Cleaning, transformation, aggregation, KPI calculation
NumPy	Numerical operations
Matplotlib	Dashboard charts and visualizations
Jupyter Notebook	Exploratory analysis and documentation
Streamlit	Interactive dashboard development
Git & GitHub	Version control and project hosting
Streamlit Community Cloud	Live dashboard deployment

16. Future Scope

- Geographic route mapping and distance-based analysis.
- Validated delivery-time and service-level analysis.
- Shipping-cost and route-profitability analysis.
- Predictive shipping-delay modeling after source-date validation.
- Automated KPI monitoring and anomaly detection.
- Integration with validated near-real-time logistics data.

17. Conclusion

The final Nassau Candy Shipping Route Efficiency project now combines exploratory analysis with a deployed interactive dashboard. It identifies Pacific as the leading region, Standard Class as the dominant mode, and Chocolate -> California as the largest route. The updated dashboard improves interpretability through KPI context, rankings, percentages, trend charts, filter consistency, and clearer visual hierarchy.

The main analytical limitation remains the source-date issue behind the unusually high calculated lead times. With that limitation clearly documented, the project is suitable as a portfolio demonstration of end-to-end data analysis, dashboard development, business interpretation, version control, and cloud deployment.

18. Project Information

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